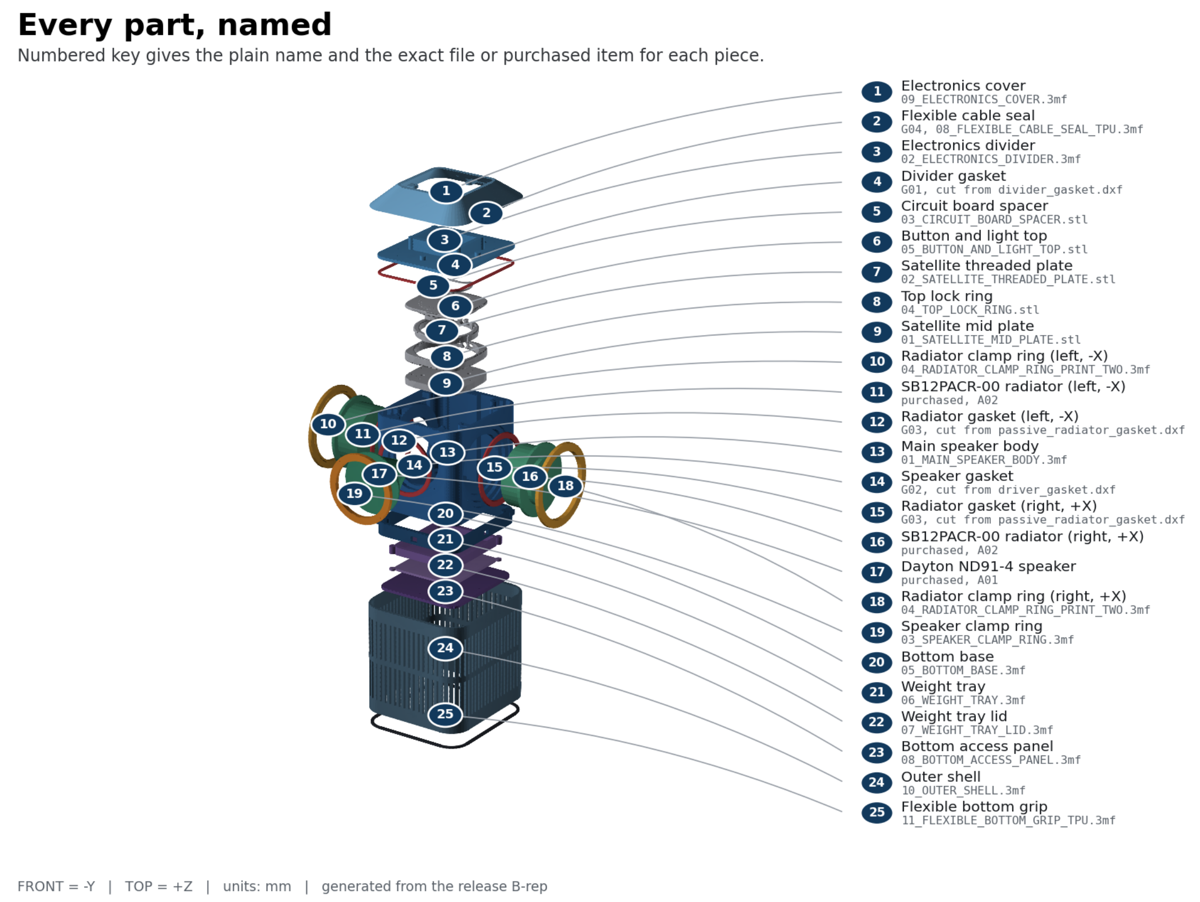
Illustrated Assembly Guide

Front is -Y, rear is +Y, left/right radiators are -X/+X, and +Z points toward the microphones. All torque values are `ENGINEERING_ESTIMATE` until the selected insert/process is pull-tested.

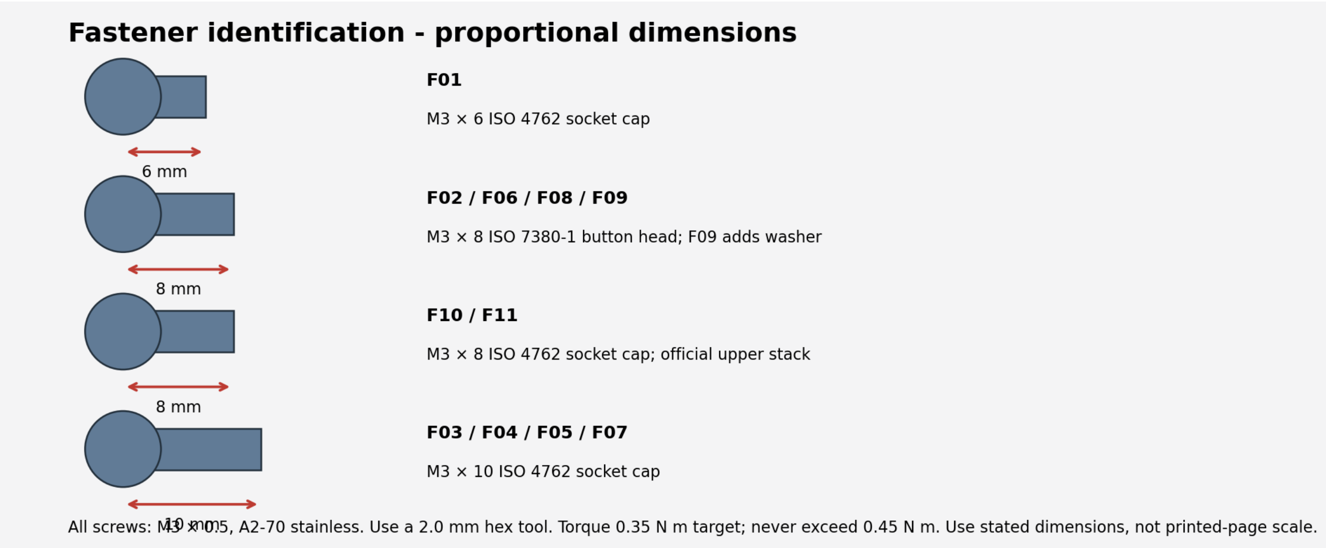


Exploded part identification

Exploded-view part key

Exact part name	Qty	Material
anti_slip_ring	1	TPU 95A
outer_shell	1	ASA
main_cabinet	1	ASA
pressure_divider	1	ASA
electronics_shroud	1	ASA
active_driver_clamp_ring	1	ASA
passive_radiator_clamp_ring	2	ASA
divider_gasket	1	2 mm closed-cell EPDM
driver_gasket	1	2 mm closed-cell EPDM
passive_radiator_gasket	2	2 mm closed-cell EPDM
cable_gland	1	TPU 95A
base_skirt	1	ASA
bottom_service_plate	1	ASA

Exact part name	Qty	Material
ballast_cartridge	1	ASA
ballast_cartridge_lid	1	ASA
official_mid_plate	1	ASA (PETG alternative)
official_mid_plate_threads	1	ASA (PETG alternative)
official_pcb_spacer	1	ASA (PETG alternative)
official_lock_ring	1	ASA (PETG alternative)
official_top_plate	1	ASA (PETG alternative)
official_top_plate_snap_in_diffuser_ring	1	ASA (PETG alternative)



Fastener identification

Step 1: Identify and inspect the hardware

• Parts: Batch 1 Core rev4.1 and HAT rev4.1; O01 official_mid_plate; O02 official_mid_plate_threads; O03 official_pcb_spacer; O04 official_lock_ring; O05 official_top_plate; O06 official_top_plate_snap_in_diffuser_ring

• Fasteners: none

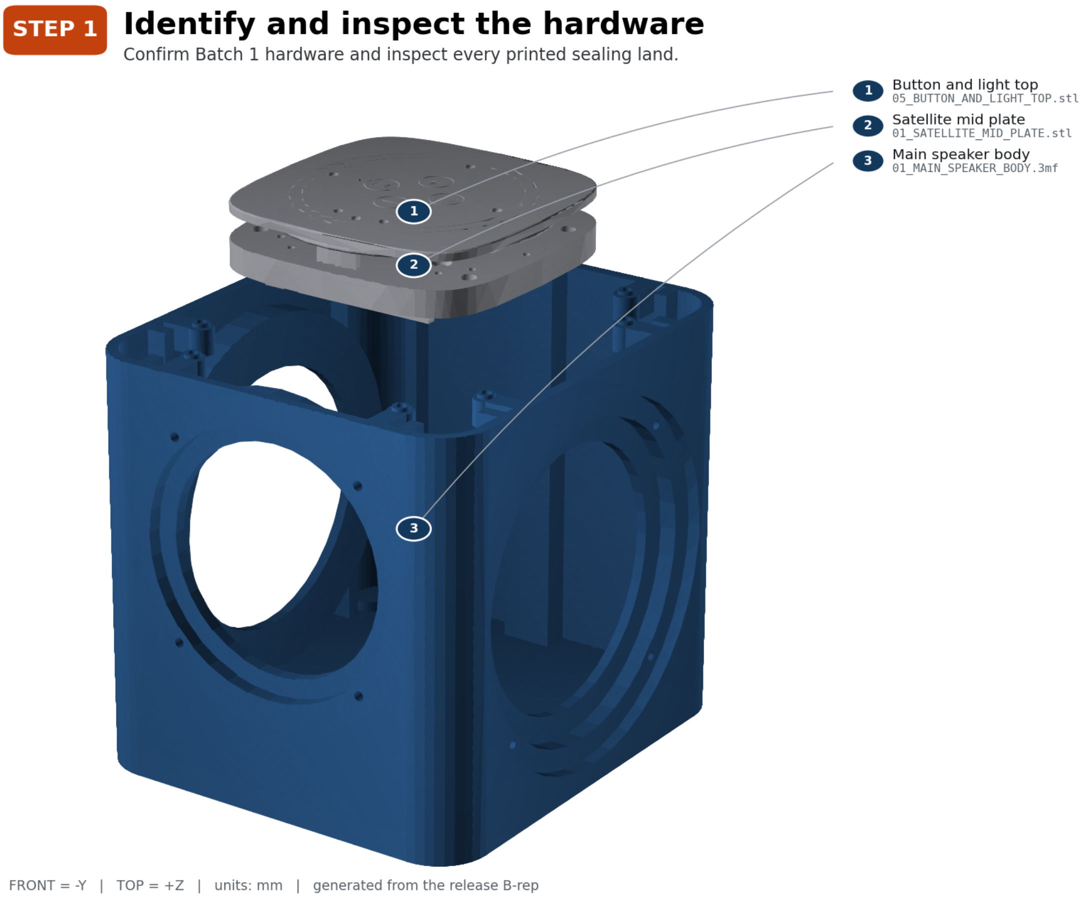
• Tools: bright light; calipers

• Gasket/seal: none

• Action: Confirm the board revision labels. Reject Batch 2 / Satellite1.1. Check off all six required official filenames in OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL and every required custom 3MF in the Printing Guide. Inspect every sealing face and remove strings without rounding an edge.

• Pass: Correct Batch 1 hardware and every required printed part are present; no crack, warp, blocked bore, or damaged gasket land.

• Warning: Do not force or approximately place the Core. Its exact stack placement requires the physical official hardware.



Step 1 - Identify and inspect the hardware

Step 2: Install and cold-check all M3 inserts

• Parts: main_cabinet, pressure_divider, base_skirt, ballast_cartridge, outer_shell

• Fasteners: H01 inserts

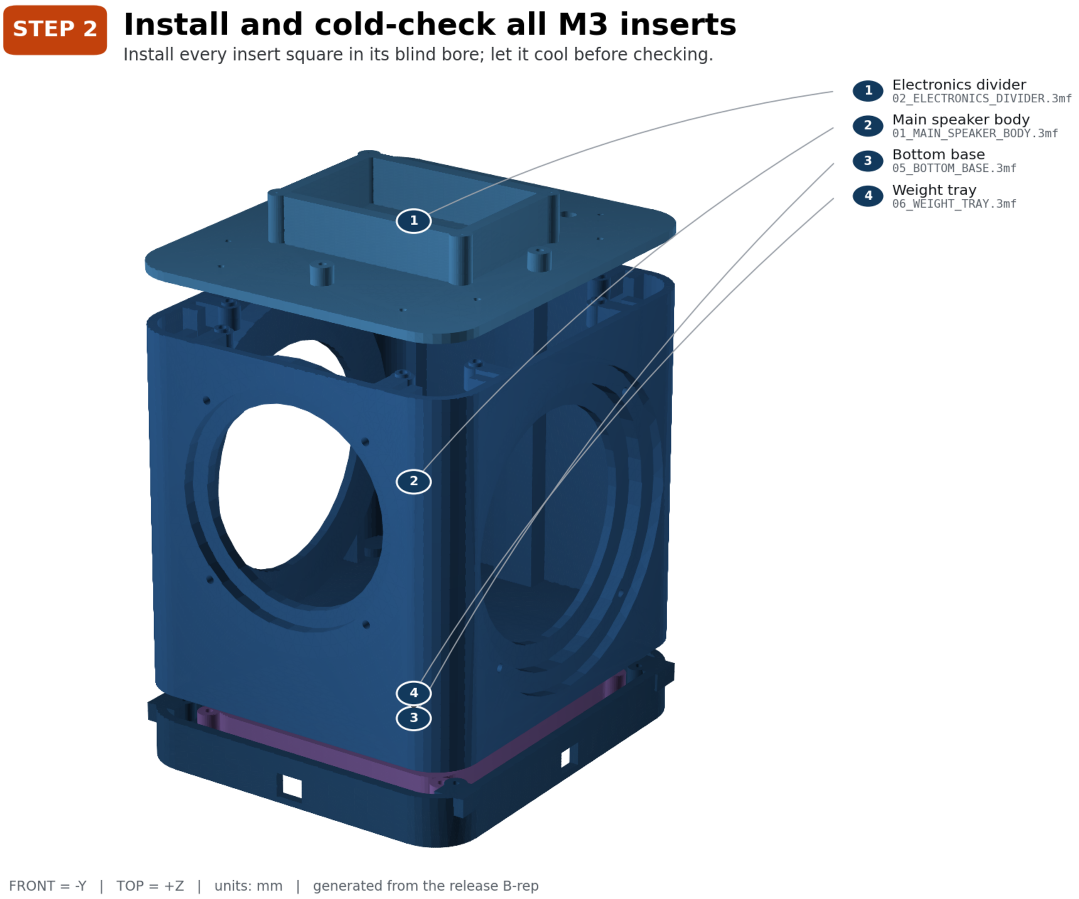
• Tools: temperature-controlled iron; M3 insert tip; square

• Gasket/seal: none

• Action: At 250-270 C, press each insert into its labeled blind bore until flush and square. Let every insert cool for five minutes. Thread an M3 screw by hand for three turns.

• Pass: No insert spins, tilts, protrudes, or blocks before three turns.

• Warning: Do not torque a hot insert. Fumes and the iron can burn; ventilate and use eye protection.



Step 2 - Install and cold-check all M3 inserts

Step 3: Wire and clamp the active driver

• Parts: main_cabinet, Dayton ND91-4, driver_gasket, active_driver_clamp_ring, JST-XH lead

• Fasteners: F04, 4 screws

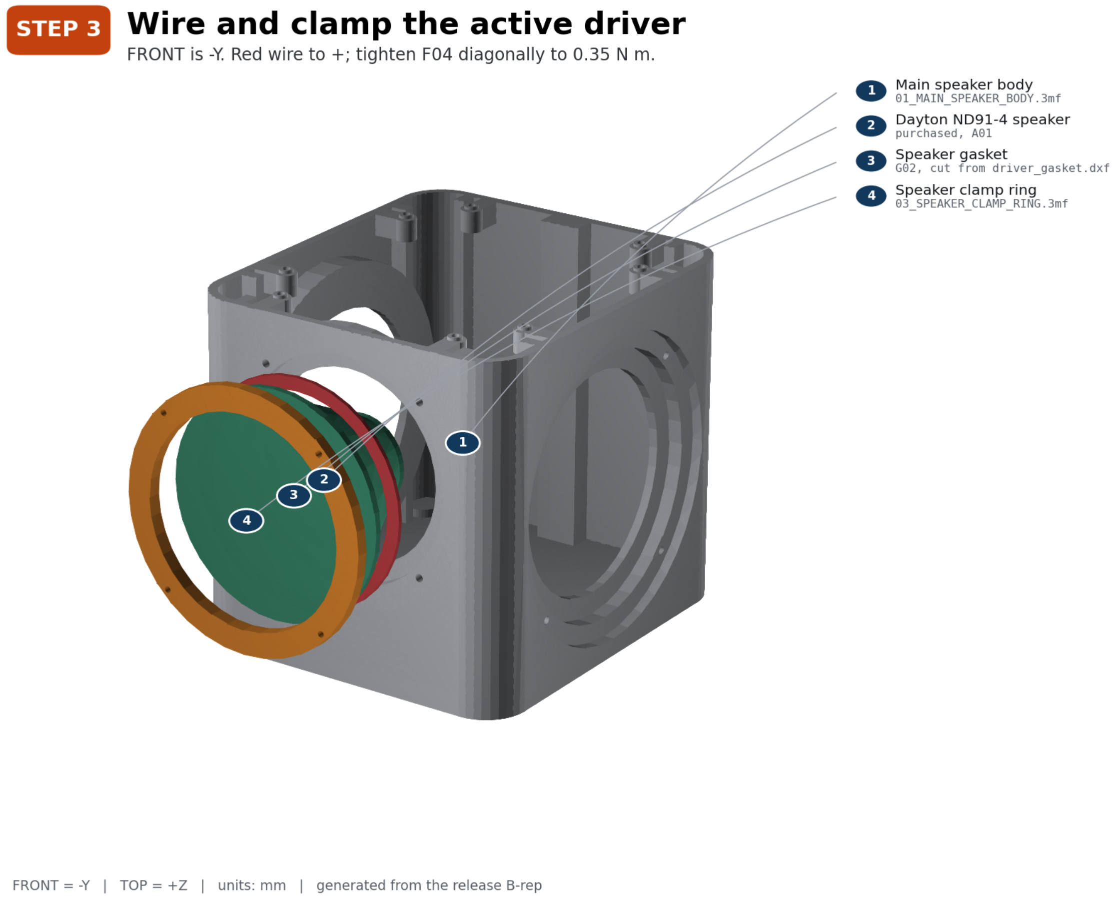
• Tools: 2.0 mm hex; crimper or soldering iron; polarity tester

• Gasket/seal: G02

• Action: Mark the red conductor positive. Connect red to the terminal marked + and black to -. Face the terminals upward. Center G02, seat the driver from the -Y/front side, fit the clamp ring, and tighten F04 in two diagonal passes to 0.35 N m; never exceed 0.45 N m.

• Pass: Ring bottoms evenly; gasket is not visible in the bore; cone moves outward on a brief 1.5 V positive polarity pulse.

• Warning: Use only a brief low-voltage polarity pulse. Never connect a loose driver to the powered HAT.



Step 3 - Wire and clamp the active driver

Step 4: Mass-match and clamp both passive radiators

• Parts: 2 SB12PACR-00, 2 passive_radiator_gaskets, 2 clamp rings, matched M6 washer stacks

• Fasteners: F05, 8 screws total

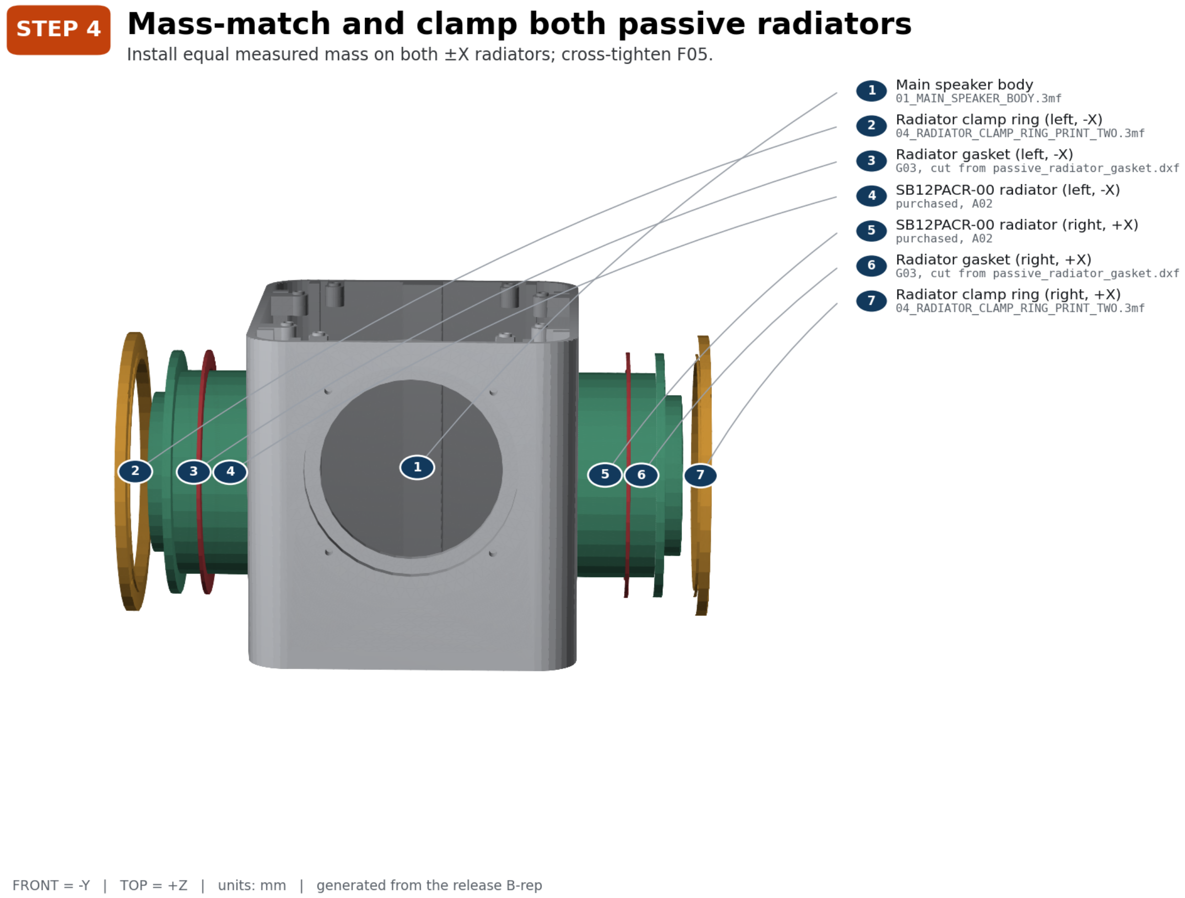
• Tools: 0.01 g scale; 2.0 mm hex

• Gasket/seal: G03, one per side

• Action: Weigh two identical tuning stacks to the value in reports/acoustics/summary.json. Secure one to each M6 post. Install radiators on +/-X with matching orientation, then tighten each F05 crosswise to 0.35 N m; never exceed 0.45 N m.

• Pass: Added masses match within 0.02 g; both rings bottom evenly; surrounds move freely and do not touch the shell keep-out.

• Warning: Unequal mass defeats reaction-force cancellation. Do not press on either cone.



Step 4 - Mass-match and clamp both passive radiators

Step 5: Route the cable, close the divider, and leak-check

• Parts: pressure_divider, divider_gasket, leak_test_adapter, cable_gland

• Fasteners: F03, 8 screws

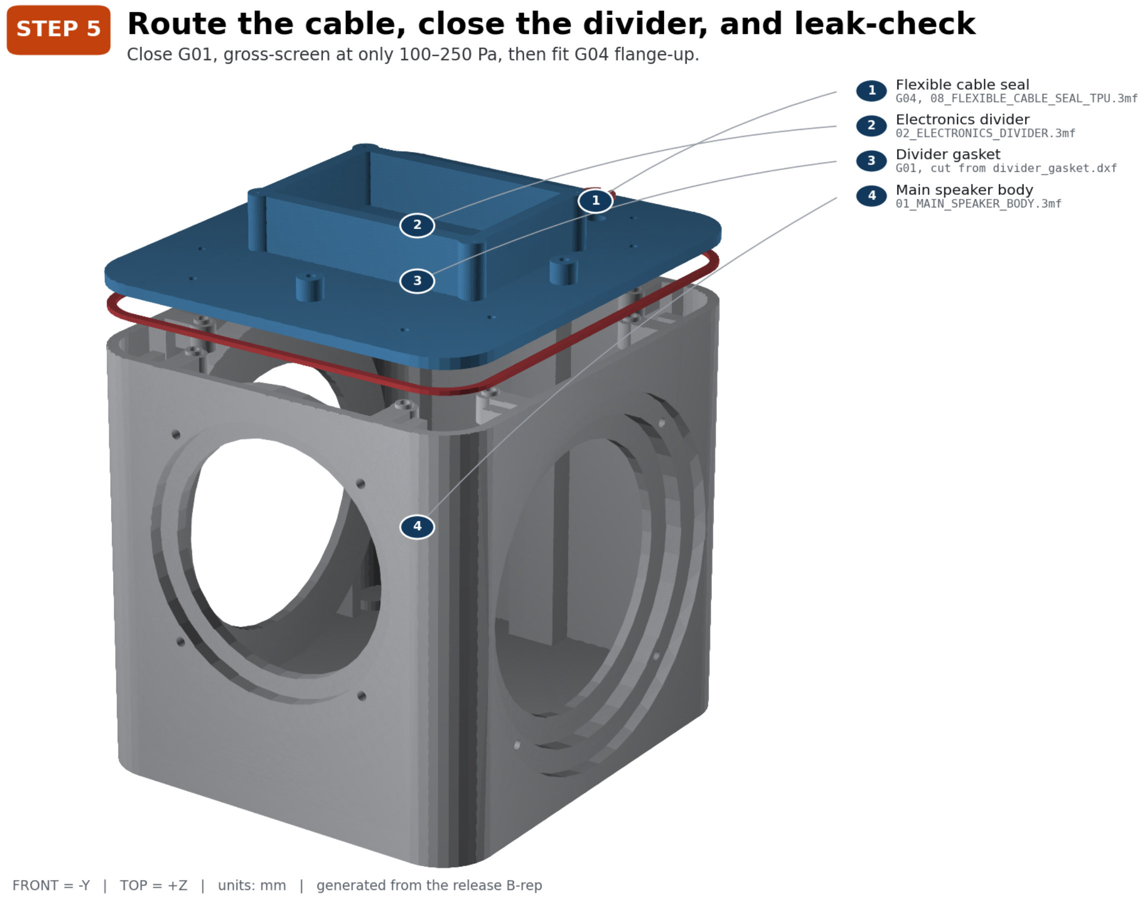
• Tools: 2.0 mm hex; hand bulb; 0-500 Pa gauge; leak-detection solution

• Gasket/seal: G01; temporary adapter then G04

• Action: Pass both conductors through the divider. Fit the temporary adapter over them, place G01 without twists, and tighten F03 in a star pattern to 0.35 N m. Apply only 100-250 Pa with a hand bulb. Brush leak solution on external gasket seams; no bubbles are allowed. Vent, pull the adapter upward, and install G04 with its flange toward the electronics bay.

• Pass: No growing bubbles, abnormal diaphragm displacement, or audible leak; final gland cannot rotate or lift by finger force.

• Warning: Never use shop air, never exceed 250 Pa, and keep liquid away from electronics. This is a gross-leak screen, not an acoustic-Q measurement.



Step 5 - Route the cable, close the divider, and leak-check

Step 6: Install the base and retained ballast

• Parts: base_skirt, ballast_cartridge, 2 steel plates, ballast lid, bottom_service_plate

• Fasteners: F06, F07, F08

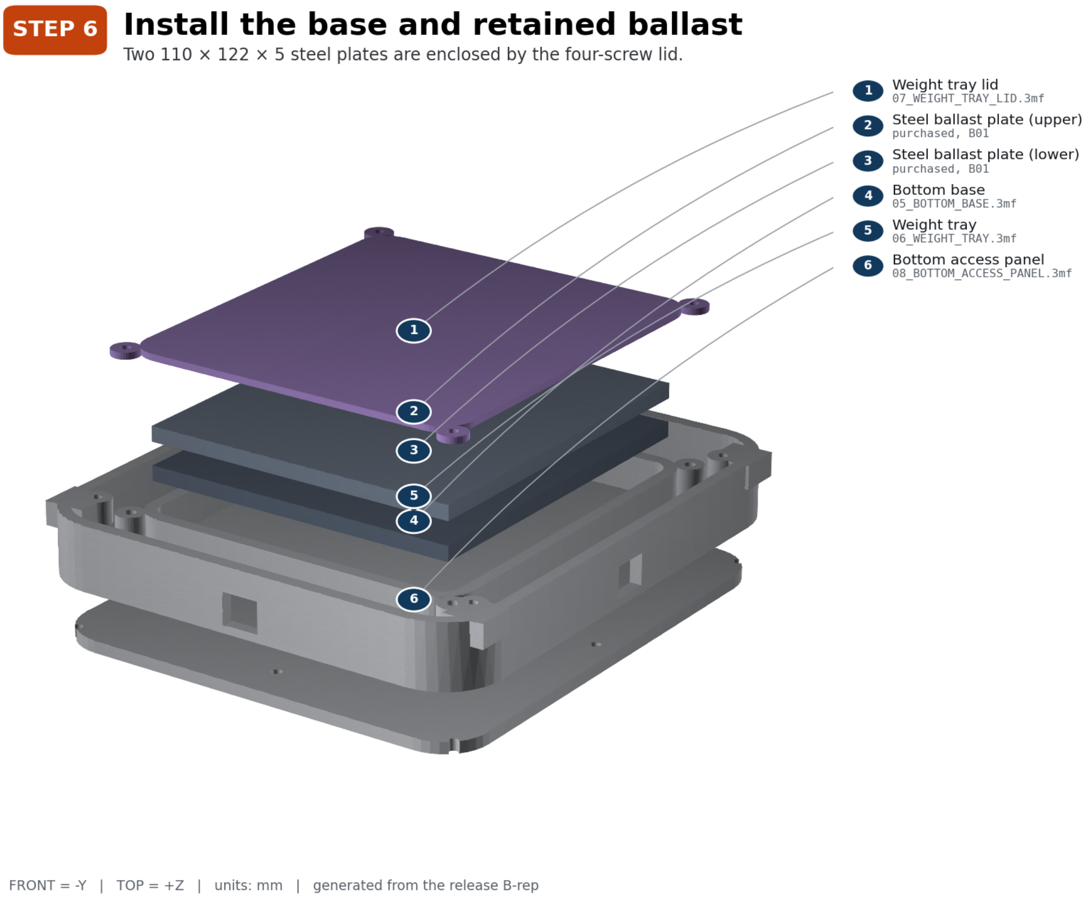
• Tools: 2.0 mm hex; scale

• Gasket/seal: none

• Action: Attach the base skirt with F07. Place both deburred dry plates flat in the cartridge; there must be no rocking. Install the lid with F06, insert the cartridge from below, and capture it with the bottom service plate using F08.

• Pass: Cartridge mass is approximately 1054 g; no plate moves when shaken gently; all four lid screws engage at least 3 mm.

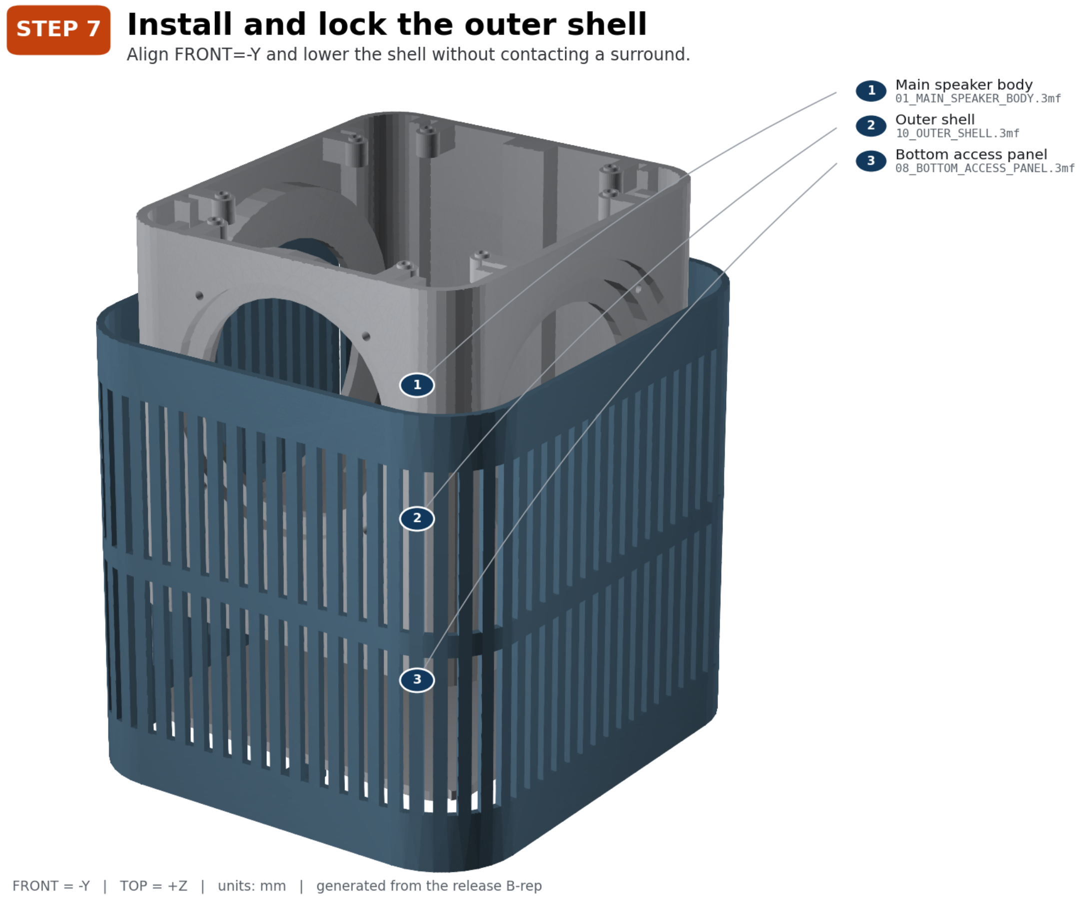
• Warning: The steel stack is heavy. Keep fingers clear and do not operate the unit without the retained lid and service plate.



Step 6 - Install the base and retained ballast

Step 7: Install and lock the outer shell

- Parts: outer_shell; lower assembly
- Fasteners: F09, 4 screws with nylon washers
- Tools: 2.0 mm hex
- Gasket/seal: none
- Action: Align FRONT with -Y and slide the shell downward without touching either surround. Invert on a soft mat and install F09 through the bottom service plate into the shell bosses.
- Pass: Every slot is clear; shell has an even reveal and at least 2 mm moving-part clearance; no wire is visible near a cone.
- Warning: Stop if the shell contacts a clamp ring or surround. Do not flex the shell over an obstruction.



Step 7 - Install and lock the outer shell

Step 8: Install the shroud and official Batch 1 upper stack

• Parts: electronics_shroud; O01-O06 official prints; Batch 1 HAT/Core

• Fasteners: F01, F02, F10, and F11; 4 of each

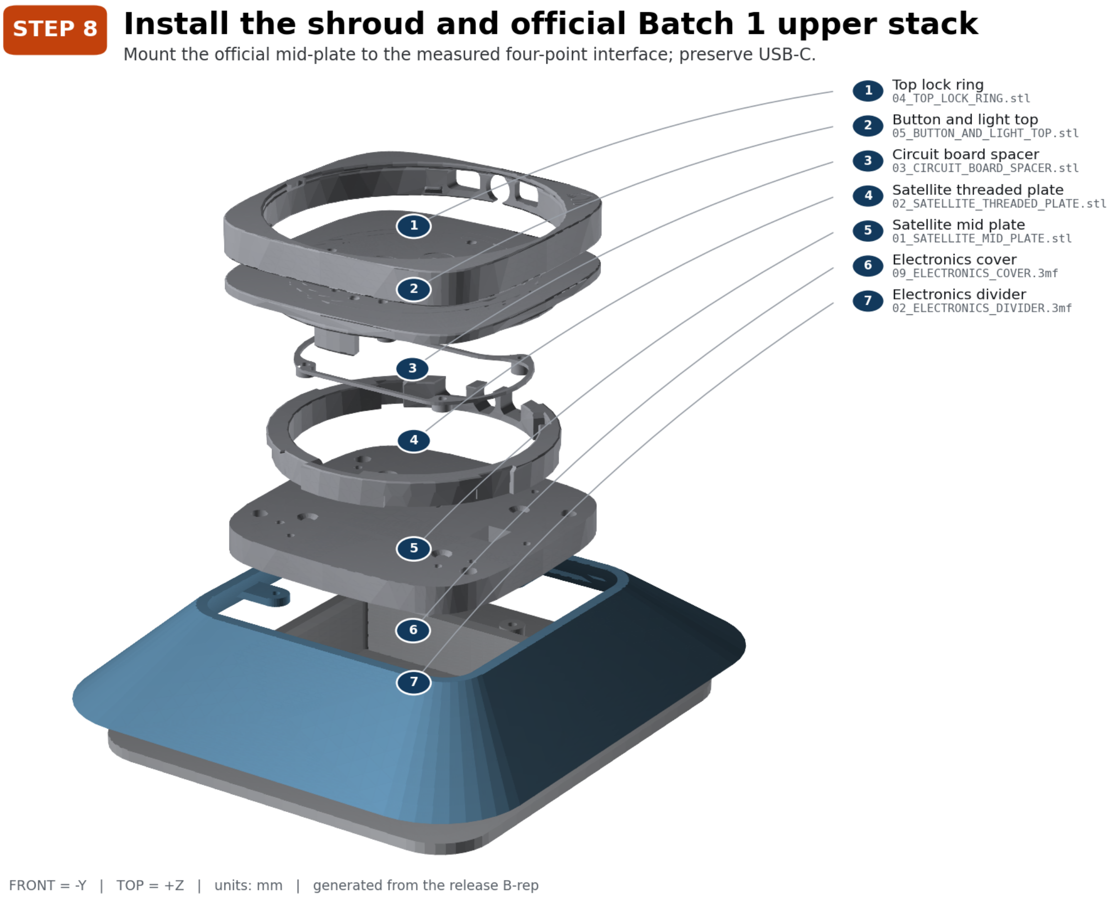
• Tools: 2.0 mm hex; ESD-safe bench

• Gasket/seal: none; electronics bay is outside the acoustic chamber

• Action: Bolt the shroud to its four outboard bosses with F02. Seat O01 on the four measured divider bosses and install F01. Snap O06 into O05 (or use both O07/O08 during a multi-material O05 print; never install O06 and O08 together). Align O03's taller standoffs with the I/O side and locate the HAT. Install the Core/HAT using the official Batch 1 sequence. Align the logos and I/O on O04/O05, engage the snaps, and rotate the lock ring. Align O02's four nubs with O01 and keep I/O toward rear/+Y. Connect the keyed JST-XH speaker plug before closure.

• Pass: Mid-plate sits on all four bosses; USB-C remains reachable; cable has service slack and cannot enter a moving-part envelope; buttons click and diffuser/LED apertures remain clear.

• Warning: Core placement is REQUIRES_PHYSICAL_VALIDATION. Follow the official Batch 1 instructions and stop at any collision; do not improvise a transform from the CAD envelope.



Step 8 - Install the shroud and official Batch 1 upper stack

Step 9: Fit the anti-slip ring and complete inspection

• Parts: anti_slip_ring; complete assembly

• Fasteners: none

• Tools: hands; flashlight

• Gasket/seal: inspect G01-G04

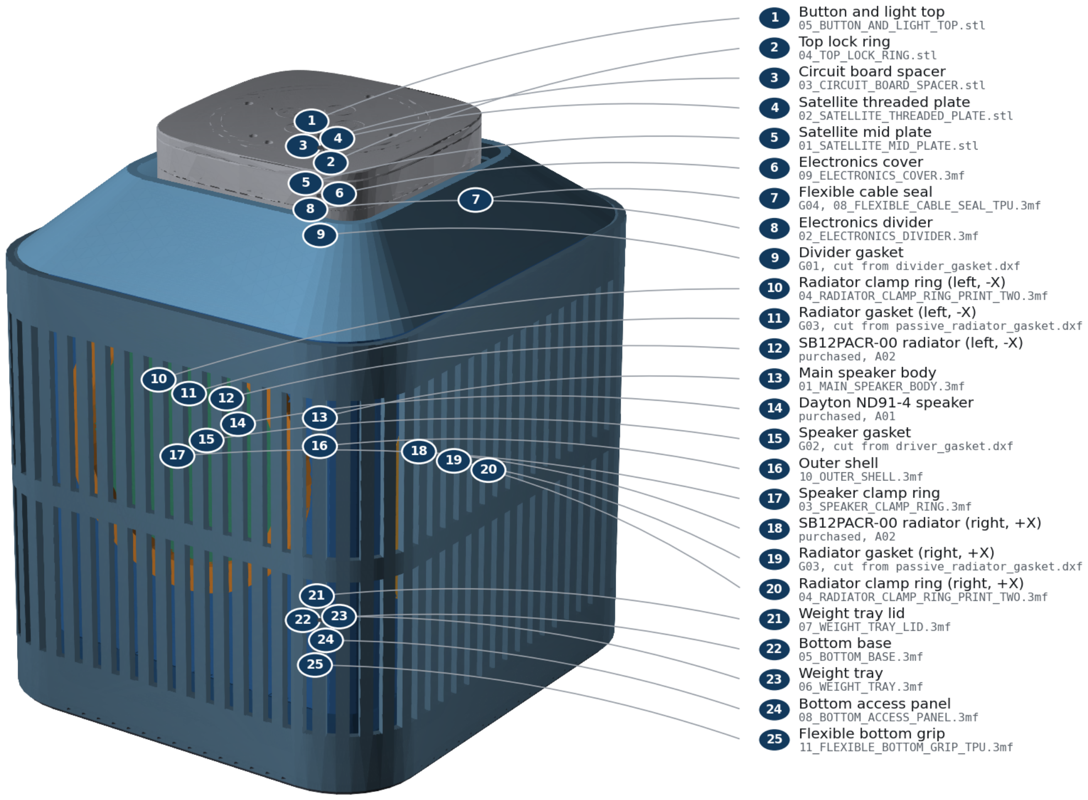
• Action: Stretch the TPU ring evenly around the bottom rim. Set the unit upright and inspect all seams, fastener heads, slots, cable exits, buttons, and moving components.

• Pass: Unit stands without rocking; no rattle is heard during gentle handling; every fastener is present and every seal is continuously compressed.

• Warning: Do not power the unit until the commissioning checklist is ready.

STEP 9 Fit the anti-slip ring and complete inspection

Final inspection: all seams even, all openings clear, no loose hardware.



FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

Step 9 - Fit the anti-slip ring and complete inspection

Source commit c83cac321c50. No physical validation has been performed.